MIL-H-3663B
15 April 1975
SUPERSEDING
MIL-H-003663A (DM)
25 October 1967 and
MIL-H-3663
8 January 1952

MILITARY SPECIFICATION

HAMMER, HAND (OPTICAL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a riveting hammer suitable for use in optical shops.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-P-40	Packaging and Packing of Hand Tools

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, and standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC-5120

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:10Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) Standard

E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Standard Methods of Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Material.

3.1.1 Head. The head shall be made of cast malleable iron, suitably treated, the round end of the head having a Rockwell hardness of C50 to C62.

3.1.2 Handle. The handle shall be made of thoroughly seasoned maple or other hardwood suitable for the purpose.

3.2 Style and design. Style and design shall conform to figure 1, and shall be a riveting type hammer, known commercially as a watchmaker's hammer.

3.2.1 Head. The head shall have a bright finish, and shall be free from irregularly finished and unfinished areas. The shape of the head shall be as shown in figure 1. The striking face shall be even and uniform across the surface, and free from irregular and unfinished areas. Corners and edges at the face end of the head shall be chamfered. The head shall be provided with an eye through which the handle shall be tightly secured.

3.2.2 Handle. Each hammer shall be fitted with a well proportioned handle of the size and shape as shown in figure 1. Handle shall fit into the eye, free of space between the eye and the handle, and shall be secured with a steel wedge or wedges. Handles shall be coated with shellac and tumbled or otherwise polished to a smooth finish.

3.2.2 Weight. The overall weight of the hammer shall be two and one half ounces, plus or minus five percent. The head shall weigh approximately two ounces.

3.3 Identification marking. Each hammer shall be legibly and permanently marked on the head with the manufacturer's name or registered trademark.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:10Z
Check the source to verify that this is the current version before use.

3.4 Workmanship. Workmanship shall be first class throughout. Hammers shall be free from burrs, sharp edges, splinters, rough surface and any other defects which detract from their appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order the supplier may use how own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, Inspection level II. AQL's shall be as specified in 4.3.1.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, Inspection level S-1, AQL shall be 4.0 (percent defective).

4.2.2.1 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The hammers shall be examined for defects, including, but not limited to those listed in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:10Z
Check the source to verify that this is the current version before use.

Table I. Classification of Defects

Categories	Defects
Major	
101	Hammer is not of the specified design.
102	Head is not tightly secured to handle.
Minor	
201	Dimension out of tolerance.
202	Sharp edges or burrs on metal.
203	Splinters and surface roughness on wood.
204	Identification marking missing.
205	Irregular and uneven striking face.
206	Finish of head not bright and uniform.
207	Wooden handle not shellacked and polished to a smooth finish.
208	Packaging, packing or marking not as specified.

4.3.1 Visual and dimensional examination. Any hammer containing one or more visual or dimensional defects shall be rejected, and if the number of defects exceeds an AQL of 6.5 the lot represented by the sample shall be rejected.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness tests. Hardness shall be conducted in accordance with ASTM E18 to determine compliance with 3.1.1. Any failure to meet the specified requirements shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation, Packaging, Packing and Marking. Preservation, packaging, packing and marking shall be as specified in PPP-P-40. The level of preservation, packaging, and packing shall be level A, B, or C, as specified (see 6.2).

6. NOTES

6.1 Intended use. The optical hammer is intended for use as an aid in the riveting of new and replacement hanges for spectacle fronts and temples.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Packaging and packing levels (see 5.1)

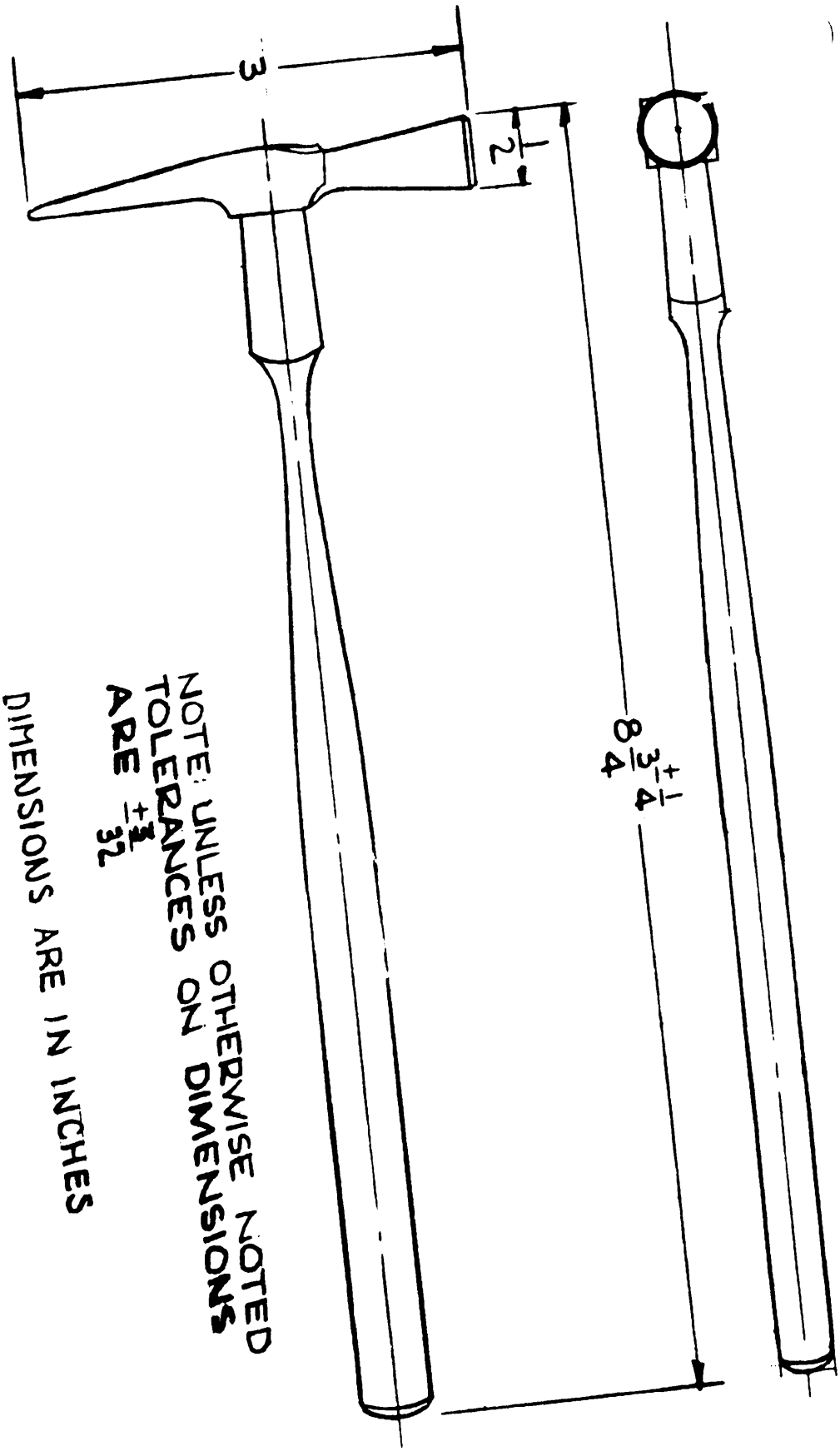
Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:10Z
Check the source to verify that this is the current version before use.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 5120-0630
is

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:10Z
Check the source to verify that this is the current version before use.



NOTE: UNLESS OTHERWISE NOTED TOLERANCES ON DIMENSIONS ARE ±1/32

DIMENSIONS ARE IN INCHES

FIGURE 1. HAMMER, OPTICAL

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-H-3663B HAMMER, HAND (OPTICAL)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:10Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, Pa. 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:10Z
Check the source to verify that this is the current version before use.